

JavaScript JSON

JSON (JavaScript Object Notation) is a lightweight, text-based format used to store and exchange data between applications. It is easy for humans to read and write, and easy for computers to parse and generate.

Points to Remember

JSON is used to store and organize data.

It helps in sending data:

Server → Client

Client → Server

Server → Server

JSON is language-independent and widely used in web applications.

It is commonly used with APIs and databases.

JSON Syntax

A JSON object consists of key-value pairs.

Correct JSON Syntax:

```
{  
  "First_Name": "Alvin",  
  "Last_Name": "Prajapati",  
  "Age": 30,  
  "City": "Mumbai"  
}
```

Rules of JSON

Data is enclosed within curly braces {}.

Keys (property names) must be enclosed in double quotes.

Key and value are separated by a colon (:).

Multiple key-value pairs are separated by commas (,).

Semicolons (;) are not used in JSON.

Example in JavaScript

```
<!DOCTYPE html>
```

```
<html>

<body>

<script>

let student = {

    "name": "Rahul",

    "age": 20,

    "course": "B.Sc IT"

};

document.write("Name: " + student.name + "<br>");

document.write("Age: " + student.age + "<br>");

document.write("Course: " + student.course);

</script>

</body>

</html>
```

Example of Converting JavaScript Object to JSON

```
let student = {

    name: "Rahul",

    age: 20,

    course: "B.Sc IT"

};

let jsonData = JSON.stringify(student);

document.write(jsonData);
```

Example of Converting JSON to JavaScript Object

```
let jsonData = '{"name":"Rahul","age":20,"course":"B.Sc IT"}';

let student = JSON.parse(jsonData);

document.write(student.name);
```

.....

The Document Object Model (DOM) is a programming interface for HTML documents. When a web page is loaded in the browser, the HTML document is converted into a Document Object. The DOM allows JavaScript to access, modify, add, or delete HTML elements dynamically.

The document object is the root object of the HTML page and is a property of the window object.

`window.document`

Methods of Document Object

Method Description

`document.write("string")` Writes the given text to the document.

`document.writeln("string")` Writes the text with a newline at the end.

`document.getElementById()` Returns the element with the specified id.

`document.getElementsByName()` Returns all elements with the specified name.

`document.getElementsByTagName()` Returns all elements with the specified tag name.

`document.getElementsByClassName()` Returns all elements with the specified class name.

1. `document.write()`

Definition:

The `document.write()` method displays text directly on the web page.

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<script>
```

```
document.write("Welcome to JavaScript!");
```

```
</script>
```

```
</body>
```

```
</html>
```

2. `document.writeln()`

Definition:

The document.writeln() method writes text with a newline after each statement.

Example

```
<!DOCTYPE html>

<html>

<body>

<script>

document.writeln("Hello");

document.writeln("Students");

</script>

</body>

</html>
```

3. Accessing Field Value using document Object

You can access the value entered by the user using the form name and input name.

Example

```
<!DOCTYPE html>

<html>

<body>

<form name="form1">

Name:

<input type="text" name="username">

<input type="button" value="Show"

onclick="showName()">

</form>

<script>

function showName()

{

    alert(document.form1.username.value);

}
```

```
</script>
```

```
</body>
```

```
</html>
```

4. document.getElementById()

Definition:

Returns the element having the specified id.

Example

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<input type="text" id="name" value="Rahul">
```

```
<button onclick="show()">Click</button>
```

```
<script>
```

```
function show()
```

```
{
```

```
    alert(document.getElementById("name").value);
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

5. document.getElementsByName()

Definition:

Returns all elements with the specified name.

Syntax

```
document.getElementsByName("name");
```

Example

```
<!DOCTYPE html>
```

```
<html>

<body>

<input type="radio" name="gender" value="Male"> Male
<input type="radio" name="gender" value="Female"> Female

<script>

let g = document.getElementsByName("gender");
console.log(g.length);

</script>

</body>

</html>
```

6. document.getElementsByTagName()

Definition:

Returns all elements with the specified HTML tag.

Syntax

```
document.getElementsByTagName("p");
```

Example

```
<!DOCTYPE html>

<html>

<body>

<p>Paragraph 1</p>
<p>Paragraph 2</p>

<script>

let p = document.getElementsByTagName("p");
alert(p.length);

</script>

</body>

</html>
```

7. document.getElementsByClassName()

Definition:

Returns all elements with the specified class name.

Example

```
<!DOCTYPE html>

<html>

<body>

<p class="demo">First</p>

<p class="demo">Second</p>

<script>

let x = document.getElementsByClassName("demo");

alert(x.length);

</script>

</body>

</html>
```

8. innerHTML

Definition:

The innerHTML property is used to read or change HTML content inside an element. It interprets HTML tags.

Example

```
<!DOCTYPE html>

<html>

<body>

<p id="demo"></p>

<script>

document.getElementById("demo").innerHTML =

"<b>Welcome to JavaScript</b>";

</script>

</body>
```

</html>

9. innerText

Definition:

The innerText property is used to read or change only text. HTML tags are displayed as plain text and are not interpreted.

Example

```
<!DOCTYPE html>
<html>
<body>
<p id="demo"></p>
<script>
document.getElementById("demo").innerText =
"<b>Welcome to JavaScript</b>";
</script>
</body>
</html>
```

Output

Difference between innerHTML and innerText

innerHTML	innerText
Displays HTML content and interprets HTML tags.	Displays only plain text.
Used for creating dynamic HTML.	Used for displaying dynamic text.
HTML tags are rendered.	HTML tags are shown as text.